

Abdullah Majali

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EDUCATION

Princess Sumaya University for Technology

BSc. Data Science and AI

Amman, Jordan

Sept 2020

EXPERIENCE

Full-stack AI/Machine Learning Engineer

Royal Hashemite Court

July 2025 – Present

Amman, Jordan

- Architected and deployed a hybrid Retrieval-Augmented Generation system combining BM25, sparse, and dense vector retrieval using Qdrant to enable low-latency, high-precision document question answering at scale.
- Built production-grade document processing pipelines for PDFs and scanned documents, performing text extraction, normalization, and structured ingestion into MongoDB and PostgreSQL.
- Integrated Vision-Language Models for scanned document understanding, implementing skew correction, image enhancement, region-of-interest detection, and fallback retry logic with adaptive resizing and field-level validation, improving extraction accuracy from 60% to 80% over baseline VLM OCR.
- Implemented automated data validation and anomaly detection pipelines that enforced business rules and triggered alerts for inconsistencies and out-of-policy values.
- Designed and implemented interactive analytics dashboards to visualize document processing outputs, validation results, and system-level metrics.
- Implemented secure, scalable document storage and delivery using an S3-compatible object store (Garage), enabling controlled access and reliable file serving.
- Containerized and deployed the full application stack using Docker and docker-compose for reproducible on-premise production environments.

Data Scientist

Eastnets

July 2024 – June 2025

Amman, Jordan

- Developed and deployed machine learning models within financial compliance and AML systems to support fraud detection, risk scoring, and alert prioritization workflows.
- Designed risk-based prioritization pipelines that ranked alerts and investigative tasks using learned risk factors, improving analyst focus on high-risk cases.
- Applied model explainability techniques including SHAP and LIME to provide transparency into model predictions and support regulatory and audit requirements.
- Designed and validated models using synthetically generated datasets to evaluate rare-event behavior and stress-test model performance under edge-case scenarios.
- Prototyped an AI copilot leveraging large language models to assist analysts with compliance workflows, integrating with existing data sources and evaluating feasibility for product integration.
- Designed and built a FastAPI-based forecasting service for a messaging warehouse product, implementing asynchronous job execution to aggregate historical message volumes from Apache Solr, train time-series models, forecast demand over user-defined horizons, and persist results with health-aware retry mechanisms.
- Built a FastAPI-based trade analysis service that extracted structured information from shipping descriptions using LLMs, mapped products to international trade codes, normalized currencies, and leveraged external web search results to flag anomalous or potentially fraudulent trade activity.

PROJECTS

BiasLens | *Python, Apache Jena, Docker*

- Designed a knowledge graph-based system for media bias detection, modeling entities and relationships using a custom ontology.
- Built a graph ML pipeline leveraging RDF data and graph embeddings to predict political bias in news articles.
- Integrated Apache Jena Fuseki as a triplestore for scalable graph querying and reasoning.
- Applied named entity recognition using GLiNER to extract entities from unstructured news text for graph population.

Argus | *Python, FastAPI, Neo4j, Celery*

- Designed a distributed pipeline that crawls academic publication metadata from OpenAlex and builds a knowledge graph of papers, authors, institutions, and topics.
- Implemented asynchronous crawl workers with rate limiting, resumable jobs, and fault-tolerant ingestion pipelines.

Minutes of Meeting | *Python, FastAPI, Qwen-ASR, LLMs*

- Built a meeting intelligence platform that transcribes and summarizes multilingual meetings using local ASR and language models.
- Implemented authentication, meeting history management, and structured summary generation.

TECHNICAL SKILLS

Languages: Python, SQL

Frameworks: Flask, FastAPI

ML: PyTorch, scikit-learn, TensorFlow, SHAP, LIME

LLMs/Retrieval: Qdrant, vLLM, llama.cpp, Ollama

Data and Infra: Docker, MongoDB, PostgreSQL, S3-compatible storage

Dev-Tools: Git, WSL2, VSCode/Cursor

PUBLICATIONS

BiasLens: Graph Machine Learning Approach for Media Bias Detection

Abdullah Majali et al.

2025 International Conference on New Trends in Computing Sciences (ICTCS) - IEEE 10.1109/ICTCS65341.2025.10989457

- Developed a graph-based machine learning framework to model and detect political bias in US news articles using structured knowledge representations and graph learning techniques.